



qwq-32b

Text Generation • Qwen

@cf/qwen/qwq-32b

QwQ is the reasoning model of the Qwen series. Compared with conventional instruction-tuned models, QwQ, which is capable of thinking and reasoning, can achieve significantly enhanced performance in downstream tasks, especially hard problems. QwQ-32B is the medium-sized reasoning model, which is capable of achieving competitive performance against state-of-the-art reasoning models, e.g., DeepSeek-R1, o1-mini.

Model Info

Context 24,000 tokens

Window [↗](#)

Model Info

LoRA	Yes
Unit Pricing	\$0.66 per M input tokens, \$1.00 per M output tokens

Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

[Launch the LLM Playground](#)

Usage

✓ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@cf/qwen/qwq-32b", {  
      messages,  
      stream: true,  
    });
```

```
    return new Response(stream, {  
      headers: { "content-type":  
"text/event-stream" },  
    });  
  },  
} satisfies ExportedHandler<Env>;
```

▼ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@cf/qwen/qwq-32b", { messages  
});  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```

▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/workers-ai/run/qwq-32b",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
```

```
https://api.cloudflare.com/client/v4/accounts/32b \
-X POST \
-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \
-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." }, { "role": "user", "content": "Why is pizza so good?" } ] }
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

• **object**

- **prompt** **string** required min 1

The input text prompt for the model to generate a response.

- **guided_json** **object**

JSON schema that should be fulfilled for the response.

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- temperature **number** default 0.15 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- top_p **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- top_k **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- seed **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- repetition_penalty **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- ▶ 1 **object**

- `messages` **array** required

An array of message objects representing the conversation history.

- `items` **object**

- `role` **string**

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- `tool_call_id` **string**

The tool call id. Must be supplied for tool calls for Mistral-3. If you don't know what to put here you can fall back to 000000001

- content **one of**

- 0 **string**

The content of the message as a string.

- 1 **array**

- items **object**

- type **string**

Type of the content provided

- text **string**

- image_url **object**

- url **string**

image uri with data (e.g.

...).

HTTP URL will not be accepted

- 2 **object**

- type **string**

Type of the content provided

- **text** **string**

- **image_url** **object**

- **url** **string**

image uri with data (e.g.

...).

HTTP URL will not be accepted

- **functions** **array**

- **items** **object**

- **name** **string** required

- **code** **string** required

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **name** **string** required

The name of the tool. More descriptive the better.

- **description** **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **1** **object**

- **type** **string** required

Specifies the type of tool (e.g., 'function').

- **function** **object** required

Details of the function tool.

- **name** **string** required

The name of the function.

- **description** **string** required

A brief description of what the function does.

- **parameters** **object** required

Schema defining the parameters accepted by the function.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **guided_json** **object**

JSON schema that should be fulfilled for the response.

- `raw` **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- `stream` **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.15 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable;

higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

Output

- **0** **object**

- **response** **string** required

The generated text response from the model

- **usage** **object**

Usage statistics for the inference request

- **prompt_tokens** **number** 0

Total number of tokens in input

- **completion_tokens** **number** 0

Total number of tokens in output

- **total_tokens** **number** 0

Total number of input and output tokens

- **tool_calls** **array**

An array of tool calls requests made during the response generation

- **items** **object**

- arguments **object**

The arguments passed to be passed to the tool call request

- name **string**

The name of the tool to be called

- ▶ 1 **string**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{  
  "type": "object",  
  "oneOf": [  
    {  
      "title": "Qwen_Qwq_32B_Prompt",  
      "properties": {  
        "prompt": {  
          "type": "string",  
          "minLength": 1,  
          "description": "The input  
text prompt for the model to generate a  
response."  
        },  
        "guided_json": {  
          "type": "object",  
          "description": "JSON  
schema that should be fulfilled for the  
response."  
        },  
        "raw": {  
          "type": "boolean",  
          "default": false,  
          "description": "If true, a  
chat template is not applied and you must
```


adhere to the specific model's expected formatting."

```
    },  
    "stream": {  
      "type": "boolean",  
      "default": false,  
      "description": "If true,  
the response will be streamed back  
incrementally using SSE, Server Sent Events."  
    },  
    "max_tokens": {  
      "type": "integer",  
      "default": 256,  
      "description": "The  
maximum number of tokens to generate in the  
response."  
    },  
    "temperature": {  
      "type": "number",  
      "default": 0.15,  
      "minimum": 0,  
      "maximum": 5,  
      "description": "Controls  
the randomness of the output; higher values  
produce more random results."
```

```
},  
"top_p": {  
  "type": "number",  
  "minimum": 0,  
  "maximum": 2,  
  "description": "Adjusts
```

the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses."

```
},  
"top_k": {  
  "type": "integer",  
  "minimum": 1,  
  "maximum": 50,  
  "description": "Limits the
```

AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises."

```
},  
"seed": {  
  "type": "integer",  
  "minimum": 1,
```

```
    "maximum": 9999999999,  
    "description": "Random  
seed for reproducibility of the generation."  
  },  
  "repetition_penalty": {  
    "type": "number",  
    "minimum": 0,  
    "maximum": 2,  
    "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."  
  },  
  "frequency_penalty": {  
    "type": "number",  
    "minimum": 0,  
    "maximum": 2,  
    "description": "Decreases  
the likelihood of the model repeating the same  
lines verbatim."  
  },  
  "presence_penalty": {  
    "type": "number",  
    "minimum": 0,  
    "maximum": 2,
```

```
      "description": "Increases
the likelihood of the model introducing new
topics."
    },
    "required": [
      "prompt"
    ]
  },
  {
    "title": "Qwen_Qwq_32B_Messages",
    "properties": {
      "messages": {
        "type": "array",
        "description": "An array
of message objects representing the
conversation history.",
        "items": {
          "type": "object",
          "properties": {
            "role": {
              "type":
"string",
              "description":
"The role of the message sender (e.g., 'user',
```

```
'assistant', 'system', 'tool')."
    },
    "tool_call_id": {
      "type":
"string",
      "description":
"The tool call id. Must be supplied for tool
calls for Mistral-3. If you don't know what to
put here you can fall back to 000000001",
      "pattern": "[a-zA-Z0-9]{9}"
    },
    "content": {
      "oneOf": [
        {
          "type": "string",
          "description": "The content of the message as
a string."
        },
        {
          "type": "array",
```

```
"items": {  
  
  "type": "object",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "Type of the content provided"  
  
    },  
  
    "text": {  
  
      "type": "string"  
  
    },  
  
    "image_url": {  
  
      "type": "object",
```

```
"properties": {  
  
  "url": {  
  
    "type": "string",  
  
    "pattern": "^data:*",  
  
    "description": "image uri with data (e.g.  
...). HTTP URL will  
not be accepted"  
  
  }  
  
}  
  
},  
  
{  
  
  "type": "object",
```

```
"properties": {  
  
  "type": {  
  
    "type": "string",  
  
    "description": "Type of the content provided"  
  },  
  
  "text": {  
  
    "type": "string"  
  },  
  
  "image_url": {  
  
    "type": "object",  
  
    "properties": {  
  
      "url": {  
  
        "type": "string",
```



```
"pattern": "^data:",  
  
"description": "image uri with data (e.g.  
...). HTTP URL will  
not be accepted"  
  
}  
  
}  
  
}  
  
}  
  
}  
  
}  
  
},  
"functions": {  
  "type": "array",  
  "items": {  
    "type": "object",  
    "properties": {  
      "name": {
```

```
        "type":  
"string"  
    },  
    "code": {  
        "type":  
"string"  
    }  
},  
"required": [  
    "name",  
    "code"  
]  
}  
,  
"tools": {  
    "type": "array",  
    "description": "A list of  
tools available for the assistant to use.",  
    "items": {  
        "type": "object",  
        "oneOf": [  
            {  
                "properties":  
  
                "name": {
```

```
"type": "string",  
  
"description": "The name of the tool. More  
descriptive the better."  
  
    },
```

```
"description": {  
  
    "type": "string",  
  
    "description": "A brief description of what  
the tool does."  
  
    },
```

```
"parameters": {  
  
    "type": "object",  
  
    "description": "Schema defining the parameters  
accepted by the tool.",  
  
    "properties": {  
  
        "type": {
```

```
"type": "string",

"description": "The type of the parameters
object (usually 'object')."

},

"required": {

"type": "array",

"description": "List of required parameter
names.",

"items": {

"type": "string"

}

},

"properties": {

"type": "object",
```

```
"description": "Definitions of each  
parameter.",  
  
"additionalProperties": {  
  
  "type": "object",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "The data type of the  
parameter."  
  
    },  
  
    "description": {  
  
      "type": "string",  
  
      "description": "A description of the expected  
parameter."
```

```
}

},

"required": [

"type",

"description"

]

}

},

"required": [

"type",

"properties"

]

}

},
```

```
        "required": [
            "name",

            "description",

            "parameters"

        ],
        {
            "properties":
            {
                "type": {

                    "type": "string",

                    "description": "Specifies the type of tool
(e.g., 'function')."

                },

                "function": {

                    "type": "object",

                    "description": "Details of the function
tool.",
```

```
"properties": {  
  
  "name": {  
  
    "type": "string",  
  
    "description": "The name of the function."  
  },  
  
  "description": {  
  
    "type": "string",  
  
    "description": "A brief description of what  
the function does."  
  },  
  
  "parameters": {  
  
    "type": "object",  
  
    "description": "Schema defining the parameters  
accepted by the function.",
```



```
"properties": {  
  
  "type": {  
  
    "type": "string",  
  
    "description": "The type of the parameters  
object (usually 'object')."  
  
  },  
  
  "required": {  
  
    "type": "array",  
  
    "description": "List of required parameter  
names.",  
  
    "items": {  
  
      "type": "string"  
  
    }  
  }  
}
```

```
},  
  
"properties": {  
  
  "type": "object",  
  
  "description": "Definitions of each  
parameter.",  
  
  "additionalProperties": {  
  
    "type": "object",  
  
    "properties": {  
  
      "type": {  
  
        "type": "string",  
  
        "description": "The data type of the  
parameter."  
  
      },  
  
    },  
  
  },  
  
}
```

```
"description": {  
  
  "type": "string",  
  
  "description": "A description of the expected  
parameter."  
  
}  
  
},  
  
"required": [  
  
  "type",  
  
  "description"  
  
]  
  
}  
  
}  
  
},
```

```
"required": [  
  
  "type",  
  
  "properties"  
  
]  
  
  },  
  
  },  
  
  "required": [  
  
    "name",  
  
    "description",  
  
    "parameters"  
  
  ]  
}  
},  
  "required": [  
    "type",  
    "function"  
  ]
```

```
    }  
  ]  
}  
},  
"guided_json": {  
  "type": "object",  
  "description": "JSON  
schema that should be fulfilled for the  
response."  
},  
"raw": {  
  "type": "boolean",  
  "default": false,  
  "description": "If true, a  
chat template is not applied and you must  
adhere to the specific model's expected  
formatting."  
},  
"stream": {  
  "type": "boolean",  
  "default": false,  
  "description": "If true,  
the response will be streamed back  
incrementally using SSE, Server Sent Events."  
},
```

```
"max_tokens": {  
  "type": "integer",  
  "default": 256,  
  "description": "The  
maximum number of tokens to generate in the  
response."  
},  
"temperature": {  
  "type": "number",  
  "default": 0.15,  
  "minimum": 0,  
  "maximum": 5,  
  "description": "Controls  
the randomness of the output; higher values  
produce more random results."  
},  
"top_p": {  
  "type": "number",  
  "minimum": 0,  
  "maximum": 2,  
  "description": "Adjusts  
the creativity of the AI's responses by  
controlling how many possible words it  
considers. Lower values make outputs more
```

predictable; higher values allow for more varied and creative responses."

```
},
```

```
"top_k": {
```

```
  "type": "integer",
```

```
  "minimum": 1,
```

```
  "maximum": 50,
```

```
  "description": "Limits the
```

AI to choose from the top 'k' most probable

words. Lower values make responses more

focused; higher values introduce more variety

and potential surprises."

```
},
```

```
"seed": {
```

```
  "type": "integer",
```

```
  "minimum": 1,
```

```
  "maximum": 9999999999,
```

```
  "description": "Random
```

seed for reproducibility of the generation."

```
},
```

```
"repetition_penalty": {
```

```
  "type": "number",
```

```
  "minimum": 0,
```

```
  "maximum": 2,
```

```
      "description": "Penalty
for repeated tokens; higher values discourage
repetition."
    },
    "frequency_penalty": {
      "type": "number",
      "minimum": 0,
      "maximum": 2,
      "description": "Decreases
the likelihood of the model repeating the same
lines verbatim."
    },
    "presence_penalty": {
      "type": "number",
      "minimum": 0,
      "maximum": 2,
      "description": "Increases
the likelihood of the model introducing new
topics."
    }
  },
  "required": [
    "messages"
  ]
}
```



```
]
}
```

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